

Finite Nagata Geometry Gives Smooth Derivative-Preserving Extensions in Separable Hilbert Space

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Status. Candidate substantial partial result for Question 1.3 of arXiv:1602.05750. Likely valid; expert review requested. The theorem below proves the full strict-derivative clause and the full global Lipschitz clause for separable real Hilbert ambient spaces. It also proves a uniform scaled-local Lipschitz clause. It does not prove the source's exact local conclusion on every $B(a, r)$ with $r < R$.

1 Source question

Koc and Kolář consider a closed set F in a normed space X , an arbitrary map $f : F \rightarrow Y$, and a Baire-one prescribed derivative $L : F \rightarrow \mathcal{L}(X, Y)$ [1]. Their Theorem 1.1 constructs an extension $\bar{f} : X \rightarrow Y$ preserving continuity, pointwise Hölder behavior, and every relative Fréchet derivative represented by L ; the extension is smooth off F whenever X has smooth partitions of unity. If $\dim X < \infty$, it also has the following two properties:

- (vii) At every $a \in F$ where L is continuous and $L(a)$ is a relative strict derivative of f , the derivatives of $\bar{f}$ off F converge to $L(a)$, and $L(a)$ is the strict derivative of $\bar{f}$ at a .
- (viii) If f is Lipschitz and L bounded on $B(a, R) \cap F$, then $\bar{f}$ is Lipschitz on $B(a, r)$ for every $r < R$; if the hypotheses hold globally on F , then $\bar{f}$ is globally Lipschitz.

Question 1.3 on page 2 asks:

Can the condition $\dim X < \infty$ in Theorem 1.1 (vii) and (viii) be replaced by the condition $\dim_N F < \infty$?

- (i) $\bar{f} = f$ on F ,
- (ii) if $a \in F$ and f is continuous at a (with respect to F), then $\bar{f}$ is continuous at a ,
- (iii) if $a \in F$, $\alpha \in (0, 1]$ and f is α -Hölder continuous at a (with respect to F), then $\bar{f}$ is α -Hölder continuous at a ; in particular, if f is Lipschitz at a (with respect to F), then $\bar{f}$ is Lipschitz at a ,
- (iv) if $a \in F$ and $L(a)$ is a relative Fréchet derivative of f at a (with respect to F), then $(\bar{f})'(a) = L(a)$,
- (v) $\bar{f}$ is continuous on $X \setminus F$,
- (vi) if X admits C^p -smooth partition of unity, then $\bar{f} \in C^p(X \setminus F, Y)$.

Moreover, if $\dim X < \infty$, then

- (vii) if $a \in F$, L is continuous at a and $L(a)$ is a relative strict derivative of f at a (with respect to F), then the Fréchet derivative $(\bar{f})'$ is continuous at a with respect to $(X \setminus F) \cup \{a\}$ and $L(a)$ is the strict derivative of $\bar{f}$ at a (with respect to X),
- (viii) if $a \in F$, $R > 0$, L is bounded on $B(a, R) \cap F$ and f is Lipschitz continuous on $B(a, R) \cap F$, then $\bar{f}$ is Lipschitz continuous on $B(a, r)$ for every $r < R$; if L is bounded on F and f is Lipschitz continuous on F , then $\bar{f}$ is Lipschitz continuous on X .

Remark 1.2.

- (a) Statement (vi) of Theorem 1.1 can be modified as follows: if X admits $\mathcal{F}$ -partition of unity where $\mathcal{F}$ is a fixed class of functions on X from Lemma 2.5, then $\bar{f}|_{X \setminus F}$ is of class $\mathcal{F}$ (where we say that $g|_U$ is of class $\mathcal{F}$, on an open set U , if for every $x \in U$, there is a neighborhood V of x such that $g|_V$ is a restriction of a function from $\mathcal{F}$).
- (b) The assumptions on partitions of unity cannot be removed from (vi), see Proposition 3.3
- (c) In (vii), if we additionally assume that $L(b)$ is a relative Fréchet derivative of f at b (with respect to F) for every $b \in U$ where U is a relative neighborhood of a in F , then $(\bar{f})'$ is continuous at a (with respect to X).
- (d) The condition $\dim X < \infty$ cannot be removed from (vii) or (viii), see Remark 4.7 for an example.

If the Nagata dimension $\dim_N F$ of F is finite (see [BB1, Definition 4.26] or [BB2, p. 13]), there is no obvious obstacle for extending with preservation of the Lipschitz property [BB2, Theorem 6.26]¹. Thus we rise a natural question:

Question 1.3. Can the condition $\dim X < \infty$ in Theorem 1.1 ((vii) and (viii)) be replaced by condition $\dim_N F < \infty$?

¹Space Y can be used as the range, see [BB2, Proposition 6.5] whose proof works for normed linear spaces.

Here finite Nagata dimension means that F satisfies Nagata(n, c) for some integer n and constant c : at every scale it has a uniformly bounded cover of scale multiplicity at most $n + 1$. We use Basso's normalization of this condition [2].

2 Claimed contribution

All Hilbert spaces below are real. Balls are open.

Theorem 1 (Hilbert–Nagata extension theorem). *Let H be a separable Hilbert space, let $\emptyset \neq F \subset H$ be closed and satisfy Nagata(n, c), let Y be a normed linear space, let $f : F \rightarrow Y$ be arbitrary, and let $L : F \rightarrow \mathcal{L}(H, Y)$ be Baire one. There is an extension $\bar{f} : H \rightarrow Y$ with all of the following properties.*

- (a) $\bar{f} = f$ on F and $\bar{f} \in C^\infty(H \setminus F, Y)$.
- (b) At every $a \in F$, continuity and pointwise α -Hölder continuity of f at a are inherited by $\bar{f}$, for $0 < \alpha \leq 1$.

(c) If $L(a)$ is a relative Fréchet derivative of f at a , then $\bar{f}$ is Fréchet differentiable at a and $D\bar{f}(a) = L(a)$.

(d) If L is continuous at $a \in F$ and $L(a)$ is a relative strict derivative of f at a , then

$$D\bar{f}(x) \longrightarrow L(a) \quad (x \rightarrow a, x \in H \setminus F),$$

and $L(a)$ is the strict derivative of $\bar{f}$ at a with respect to H .

(e) If f is globally Lipschitz on F and L is bounded on F , then $\bar{f}$ is globally Lipschitz on H .

(f) There is $\theta = \theta(n, c) \in (0, 1)$ such that, whenever $a \in F$, $R > 0$, f is Lipschitz on $B(a, R) \cap F$, and L is bounded there, the extension $\bar{f}$ is Lipschitz on $B(a, \theta R)$.

Thus part (d) is the full conclusion of source item (vii), and part (e) is the global half of source item (viii). Part (f) is weaker than the local half of (viii), but its radius loss is uniform and depends only on the Nagata data.

3 Proof intuition

Finite dimensionality enters Koc–Kolář’s proof through a smooth Whitney partition with bounded overlap and

$$\sum_i \|D\phi_i(x)\| \lesssim \text{dist}(x, F)^{-1}.$$

Finite Nagata dimension supplies the bounded-overlap Whitney cover even in an infinite-dimensional ambient metric space [2, Proposition 3.1]. The missing operation is to smooth its distance cutoffs without destroying their Lipschitz constants. On a separable Hilbert space this is supplied by the smooth Lipschitz approximation theorem of Azagra, Ferrera, López-Mesas, and Rangel [3, Theorem 1]. A positive-depth cutoff also turns pointwise bounded multiplicity into genuine local finiteness.

We then use the same affine jets as the source. A Baire-one extension $A : H \setminus F \rightarrow \mathcal{L}(H, Y)$ of L is sampled at Whitney centers z_i , and

$$\bar{f}(x) = \sum_i \phi_i(x) [f(a_i) + A(z_i)(x - a_i)].$$

Near a strict point, the jet discrepancies are $o(\text{dist}(x, F))$; the partition derivative costs precisely $O(\text{dist}(x, F)^{-1})$, so $D\bar{f}(x) \rightarrow L(a)$. Bounded data give a uniform derivative bound off F . A one-dimensional lemma along line segments glues that bound to the corresponding Lipschitz estimate on the portions of the segment lying in F .

4 A smooth finite-Nagata Whitney partition

Lemma 1. *Under the hypotheses on H and F in Theorem 1, there are an index set I , points $z_i \in H \setminus F$, anchors $a_i \in F$, and a locally finite C^∞ partition of unity $(\phi_i)_{i \in I}$ on $H \setminus F$ such that, for constants*

M, C_0, C_1 depending only on (n, c) , if $d = \text{dist}(x, F)$ and $\phi_i(x) \neq 0$, then

$$\#\{i : \phi_i(x) \neq 0\} \leq M, \quad (1)$$

$$C_0^{-1}d \leq \text{dist}(z_i, F) \leq C_0d, \quad (2)$$

$$\|x - z_i\| + \|x - a_i\| \leq C_0d, \quad (3)$$

$$\sum_i \|D\phi_i(x)\| \leq \frac{C_1}{d}. \quad (4)$$

Proof. Apply Proposition 3.1 of Basso [2] with fixed numerical parameter $q > 1$. It gives a cover $(B_i)_{i \in I}$ of $H \setminus F$. Put $r_i = \text{dist}(B_i, F)$. Basso's neighborhoods $N_{\delta r_i}(B_i)$ have bounded multiplicity, where $\delta > 0$ depends only on the chosen q . To avoid any convention about open versus closed neighborhoods, set

$$U_i = \{x : \text{dist}(x, B_i) < \delta r_i/2\}.$$

There are constants α, β depending only on (c, q) such that

$$\text{diam } B_i \leq \alpha r_i, \quad \sup_{b \in B_i} \text{dist}(b, F) \leq \beta r_i,$$

Every point belongs to at most $M = 3(n+1)$ of the sets U_i .

Choose $z_i \in B_i$ and then $a_i \in F$ with $\|z_i - a_i\| \leq (\beta + 1)r_i$. If $x \in U_i$, then

$$(1 - \delta/2)r_i \leq \text{dist}(x, F) \leq (\alpha + \beta + \delta/2)r_i, \quad \|x - z_i\| + \|x - a_i\| \leq Cr_i.$$

These elementary inequalities already give (2)–(3) for functions supported in U_i .

Let

$$g_i(x) = \text{dist}(x, H \setminus U_i), \quad \varepsilon_i = \frac{\delta r_i}{32}.$$

The function g_i is 1-Lipschitz, vanishes outside U_i , and is at least $\delta r_i/2$ on B_i . By Theorem 1 of Azagra–Ferrera–López–Mesas–Rangel [3], there is a C^∞ function $s_i : H \rightarrow \mathbb{R}$ such that

$$|s_i - g_i| < \varepsilon_i, \quad \text{Lip}(s_i) \leq 2.$$

Fix $\eta \in C^\infty(\mathbb{R}, [0, 1])$ with $\eta = 0$ on $(-\infty, 2]$ and $\eta = 1$ on $[4, \infty)$, and set

$$h_i(x) = \eta(s_i(x)/\varepsilon_i).$$

Then $h_i = 1$ on B_i ,

$$\text{supp } h_i \subset \{g_i \geq \varepsilon_i\} \subset U_i, \quad \|Dh_i\| \leq \frac{C}{r_i}.$$

The supports just constructed are locally finite, a point that is stronger than mere point-finiteness. Otherwise distinct supports would meet points $y_k \rightarrow x \in H \setminus F$. By the preceding scale comparison, their scales r_{i_k} are bounded above and below by positive multiples of $\text{dist}(x, F)$. Since $g_{i_k}(y_k) \geq \varepsilon_{i_k}$ and g_{i_k} is 1-Lipschitz, for all large k one would have $g_{i_k}(x) > 0$, hence $x \in U_{i_k}$. This contradicts the multiplicity bound M .

The locally finite sum $S = \sum_i h_i$ satisfies $S \geq 1$, because the sets B_i cover and $h_i = 1$ on B_i . Define $\phi_i = h_i/S$. At most M terms are active, and direct differentiation gives

$$\sum_i \|D\phi_i(x)\| \leq 2 \sum_{i: h_i(x) \neq 0} \|Dh_i(x)\| \leq \frac{C'}{\text{dist}(x, F)},$$

using the scale comparison and the derivative bound above. This proves the lemma. $\square$

5 The affine-jet extension

We use the following source theorem. Theorem 1.6 of Koc–Kolář [1] says that for a Baire-one map $L : F \rightarrow \mathcal{L}(H, Y)$ there is a continuous map $A : H \setminus F \rightarrow \mathcal{L}(H, Y)$ such that, for every $a \in \partial F$,

$$\|A(x) - L(a)\| \frac{\text{dist}(x, F)}{\|x - a\|} \longrightarrow 0 \quad (x \rightarrow a, x \notin F), \quad (5)$$

such that $A(x) \rightarrow L(a)$ whenever L is continuous at a , and such that

$$L \text{ bounded on } B(a, 12r) \cap F \implies A \text{ bounded on } B(a, r) \setminus F. \quad (6)$$

Take A from that theorem and the data from Lemma 1. Put

$$P_i(x) = f(a_i) + A(z_i)(x - a_i), \quad \bar{f}(x) = \begin{cases} f(x), & x \in F, \\ \sum_i \phi_i(x) P_i(x), & x \notin F. \end{cases} \quad (7)$$

The sum is locally finite, so $\bar{f}$ is C^∞ on $H \setminus F$.

We record the boundary estimates, both to establish parts (a)–(c) and for later use. If $x \rightarrow a \in F$ and $\phi_i(x) \neq 0$, then (3) gives

$$a_i, z_i \rightarrow a, \quad \|x - a_i\| = O(\text{dist}(x, F)), \quad \|z_i - a\| = O(\|x - a\|). \quad (8)$$

Moreover, (2), (5), and (8) imply

$$\|(A(z_i) - L(a))(x - a_i)\| = o(\|x - a\|) \quad (9)$$

uniformly over active indices. Uniformity follows because the active anchor and center lie in fixed constant multiples of the shrinking ball about a .

If f is continuous at a , equations (7)–(9) show $\bar{f}(x) \rightarrow f(a)$. If f is pointwise α -Hölder at a , then $\|f(a_i) - f(a)\| = O(\|x - a\|^\alpha)$; the $L(a)$ term in $A(z_i)(x - a_i)$ is $O(\|x - a\|)$, and the remaining term is $o(\|x - a\|)$, proving the same pointwise α -Hölder bound for $\bar{f}$.

Finally, if $L(a)$ is a relative Fréchet derivative of f at a , then

$$\begin{aligned} \bar{f}(x) - f(a) - L(a)(x - a) &= \sum_i \phi_i(x) [f(a_i) - f(a) - L(a)(a_i - a) \\ &\quad + (A(z_i) - L(a))(x - a_i)]. \end{aligned}$$

The first bracket is $o(\|x - a\|)$ by relative differentiability and (8), and the second is $o(\|x - a\|)$ by (9). Hence $D\bar{f}(a) = L(a)$. This proves parts (a)–(c) of Theorem 1.

6 Strict differentiability

For $x \notin F$, differentiating (7) and using $\sum_i D\phi_i(x) = 0$ gives, for any affine map $Q : H \rightarrow Y$,

$$D\bar{f}(x) = \sum_i \phi_i(x) A(z_i) + \sum_i D\phi_i(x) \otimes (P_i(x) - Q(x)), \quad (10)$$

where $(\lambda \otimes y)(v) = \lambda(v)y$.

Suppose L is continuous at $a \in F$ and $L(a)$ is a relative strict derivative of f at a . The interior case is immediate, so assume $a \in \partial F$. For each $x \notin F$, choose $b_x \in F$ with

$$\|x - b_x\| < 2 \operatorname{dist}(x, F). \quad (11)$$

No nearest-point projection is being assumed. In (10) choose $Q(t) = f(b_x) + L(a)(t - b_x)$. Then

$$\begin{aligned} P_i(x) - Q(x) &= f(a_i) - f(b_x) - L(a)(a_i - b_x) \\ &\quad + (A(z_i) - L(a))(x - a_i). \end{aligned} \quad (12)$$

As $x \rightarrow a$, all active a_i, z_i, b_x tend to a . If $d = \operatorname{dist}(x, F)$, strict differentiability and the geometry lemma make the first line of (12) equal to $o(d)$, uniformly over active i . Continuity of L and property (C) of A give $A(z_i) - L(a) = o(1)$, so the second line is also $o(d)$. Combining (10), (4), and (12) yields

$$\|D\bar{f}(x) - L(a)\| \rightarrow 0. \quad (13)$$

It remains to upgrade derivative convergence off F to a strict derivative across arbitrary pairs. We isolate the elementary device.

Lemma 2 (segment gluing). *Let $u : [0, 1] \rightarrow Y$ be continuous and let $E \subset [0, 1]$ be closed. Suppose $u|_E$ is K -Lipschitz and that u is differentiable on $[0, 1] \setminus E$ with $\|u'(t)\| \leq K$. Then u is K -Lipschitz.*

Proof. On every component interval of $[0, 1] \setminus E$, scalarization by $y^* \in Y^*$ and the real mean-value theorem give the K -Lipschitz estimate; Hahn–Banach recovers the norm estimate. Given $s < t$, if $E \cap [s, t] = \emptyset$ we are done. Otherwise let p and q be its first and last points. Use the component estimate from s to p and from q to t , the hypothesis on E from p to q , and add the three bounds. $\square$

Apply the lemma to

$$u(t) = \bar{f}(x + t(y - x)) - L(a)(x + t(y - x))$$

and $E = \{t : x + t(y - x) \in F\}$. By (13), for every $\varepsilon > 0$ the derivative of u has norm at most $\varepsilon\|y - x\|$ off E , provided x, y are sufficiently close to a . Relative strict differentiability gives the same Lipschitz bound on E . Lemma 2 therefore gives

$$\|\bar{f}(y) - \bar{f}(x) - L(a)(y - x)\| \leq \varepsilon\|y - x\|.$$

This proves part (d) of Theorem 1.

7 Global and scaled-local Lipschitz bounds

Assume first that f is globally K -Lipschitz on F and that L is bounded there. Taking $r = \infty$ in (6), the operator field A is globally bounded, say $\|A\| \leq B$. For $x \notin F$, choose b_x as in (11) and use $Q \equiv f(b_x)$ in (10). The geometry lemma gives

$$\|P_i(x) - f(b_x)\| \leq K\|a_i - b_x\| + B\|x - a_i\| \leq C(K + B) \operatorname{dist}(x, F).$$

Together with (4), this yields

$$\sup_{x \in H \setminus F} \|D\bar{f}(x)\| \leq C'(K + B) < \infty. \quad (14)$$

For any line segment in H , apply Lemma 2 with the closed subset where the segment meets F . On that subset $\bar{f} = f$ is K -Lipschitz, and on its complement (14) bounds the derivative. Thus $\bar{f}$ is globally Lipschitz, proving part (e).

Now let the hypotheses hold only on $B(a, R) \cap F$. Property (6) bounds A on $B(a, R/12) \setminus F$. Choose $\theta = \theta(n, c) > 0$ small enough that for every $x \in B(a, \theta R) \setminus F$ and every active i ,

$$z_i \in B(a, R/12), \quad a_i, b_x \in B(a, R).$$

Such a choice follows directly from (3) and (11), since $\text{dist}(x, F) \leq \|x - a\|$. The preceding derivative estimate now applies on $B(a, \theta R) \setminus F$, with the local Lipschitz constant of f and the local bound for A . Every segment between two points of $B(a, \theta R)$ stays in that ball; on its intersection with F , the local Lipschitz hypothesis applies. A final use of Lemma 2 proves part (f).

8 Scope, obstruction, and novelty check

The proof uses separability only through the quantitative C^∞ smooth approximation theorem [3]. The Nagata cover itself works in an arbitrary metric ambient space. A comparable smooth approximation theorem would therefore extend the argument to other Banach classes.

The exact local clause from the source remains open in this packet. The finite-dimensional proof first establishes a fixed-radius estimate and then uses compactness to pass from local bounds to every smaller concentric ball [1, proof of Theorem 4.1]. In infinite-dimensional Hilbert space, $\overline{B(a, r)}$ is not compact. Continuity of the auxiliary field A does not bound it on the noncompact part of $B(a, r)$ lying a positive distance from F . The present argument avoids this only by shrinking far enough that property (B) supplies a uniform bound. Closing the radius gap appears to require either an exact-local boundedness refinement of the Baire-one extender or a new jet selection in the deep interior.

A bounded novelty search on 11 August 2026 covered the four lightweight run indexes and external searches for the exact title with “Nagata dimension”, fragments of Question 1.3, “strict derivative finite Nagata”, and “Whitney extension Hilbert finite Nagata”. No exact answer to Question 1.3 or this separable-Hilbert subcase was found. Basso’s finite-Nagata Whitney cover and the Azagra–Ferrera–López-Mesas–Rangel smooth approximation theorem are proof inputs; neither source identifies the theorem above.

Human-review recommendation. Review as a substantial partial solution. The primary audit points are Lemma 1’s positive- depth local-finiteness argument, the uniform use of the non-tangential estimate in (9), and segment gluing in the strict-derivative proof.

References

- [1] M. Koc and J. Kolář, *Extensions of vector-valued functions with preservation of derivatives*, arXiv:1602.05750 (2016), especially Theorem 1.1, Question 1.3, and Theorem 1.6.
- [2] G. Basso, *Lipschitz extension theorems with explicit constants*, arXiv:2310.13554, version 3 (2024), especially Proposition 3.1.
- [3] D. Azagra, J. Ferrera, F. López-Mesas, and Y. Rangel, *Smooth approximation of Lipschitz functions on Riemannian manifolds*, arXiv:math/0602051 (2006), especially Theorem 1.